Technical Lab Report

Implementation and Verification of Extended Access Control Lists (ACLs)

Target Device: Router R1
Software: Cisco Packet Tracer
Classification: Confidential - Internal
Date: July 14, 2026

1. Executive Summary

This technical report documents the configuration, deployment, and validation of Extended Access Control Lists (ACLs) implemented on the border router **R1**. The objective of this deployment is to enforce strict access control policies on network traffic crossing various subnets, preventing unauthorized communications to internal PCs, restricting local DNS lookups, and block-listing unauthorized web services on target application servers.

2. Network Security Policies

The requirements are systematically translated into three designated ACL profiles as outlined below:

Policy Objective	Source Subnet	Destination Host / Service	Protocol / Port	ACL Name / ID	Interface & Direction
Restrict LAN-to-PC1 communication	`172.16.2.0/24`	`172.16.1.1` (PC1)	Any IP traffic	**101** (Numbered)	GigabitEthernet0/1 (Inbound)
Restrict DNS service lookups	`172.16.1.0/24`	`192.168.1.100` (SRV1)	UDP / TCP 53 (Domain)	**to-serv1** (Named)	GigabitEthernet0/0 (Inbound)
Restrict HTTP & HTTPS traffic	`172.16.2.0/24`	`192.168.2.100` (SRV2)	TCP 80 (HTTP) & 443 (HTTPS)	**block-https-serv2** (Named)	Serial0/0/0 (Outbound)

3. Configuration Commands Applied to Router R1

3.1. Standard Numbered Extended Access-List (ACL 101)

Implemented on `Gig0/1` to prevent the 172.16.2.0/24 subnet from communicating with PC1 (172.16.1.1), while permitting all other IP traffic.

```
R1# configure terminal
R1(config)# access-list 101 deny ip 172.16.2.0 0.0.0.255 host 172.16.1.1
R1(config)# access-list 101 permit ip any any
R1(config)# interface GigabitEthernet0/1
R1(config-if)# ip access-group 101 in
R1(config-if)# exit
```

3.2. Named Extended Access-List to DNS Server (`to-serv1`)

Applied on `Gig0/0` in the inbound direction. It isolates the 172.16.1.0/24 subnet from issuing DNS lookups to SRV1 (192.168.1.100).

```
R1(config)# ip access-list extended to-serv1
R1(config-ext-nacl)# deny udp 172.16.1.0 0.0.0.255 host 192.168.1.100 eq domain
R1(config-ext-nacl)# deny tcp 172.16.1.0 0.0.0.255 host 192.168.1.100 eq domain
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)# exit
R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip access-group to-serv1 in
R1(config-if)# exit
```

4. Modified ACL & Resequencing Execution

Special Task Requirements:
We target the named ACL `block-https-serv2` to drop an obsolete rule allowing HTTP block-listing for 172.16.1.0 (previously assigned sequence 10: `deny tcp 172.16.1.0 0.0.0.255 host 192.168.2.100 eq 80`). Post-removal, we run the `resequence` utility to align sequence intervals.

The operational CLI inputs implemented to successfully modify, resequence, and apply the ACL are detailed below:

```
R1(config)# ip access-list extended block-https-serv2
R1(config-ext-nacl)# no 10
R1(config-ext-nacl)# exit
R1(config)# ip access-list resequence block-https-serv2 10 10
R1(config)# interface Serial0/0/0
R1(config-if)# ip access-group block-https-serv2 out
R1(config-if)# exit
```

5. Post-Deployment Verification

Running the diagnostic show command on **R1** ensures all rules are active and confirms the removal of sequence 10 from `block-https-serv2` along with proper resequencing. Notice that sequence intervals are cleanly set back to increments of 10 starting from 20.

```
R1# show access-lists
Extended IP access list 101
    10 deny ip 172.16.2.0 0.0.0.255 host 172.16.1.1 (4 matches)
    20 permit ip any any (149 matches)
Extended IP access list to-serv1
    10 deny udp 172.16.1.0 0.0.0.255 host 192.168.1.100 eq domain (4 matches)
    20 deny tcp 172.16.1.0 0.0.0.255 host 192.168.1.100 eq domain
    30 permit ip any any (4 matches)
Extended IP access list block-https-serv2
    20 deny tcp 172.16.2.0 0.0.0.255 host 192.168.2.100 eq www (57 matches)
    30 deny tcp 172.16.2.0 0.0.0.255 host 192.168.2.100 eq 443 (12 matches)
    40 permit ip any any (32 matches)
```

6. Validation Testing Results

- **Ping to PC1 Blocked:** PC3 (`172.16.2.1`) pinging PC1 (`172.16.1.1`) successfully drops with a `Destination host unreachable` notification from default gateway `172.16.2.254`, highlighting successful block-listing.
- **DNS Query Blocked:** Executing `nslookup google.com 192.168.1.100` on PC1 yields a `DNS request timed out. timeout was 15000 milli seconds`, validating strict port 53 drop.
- **HTTP/HTTPS Services Blocked:** Web browser requests to `http://192.168.2.100` from PC4 display `Request Timeout`. Conversely, other services such as FTP (port 21) allow open telnet terminal access, proving protocol-specific filtering is fully operational.

> **Conclusion:** All security policies have been implemented and verified successfully. Removing the individual rule and subsequently executing the resequencing command cleaned up the configuration table without interrupting routing capabilities.